

Tilsynsapp-PNB — Executive Summary

A digital tool for efficient building site visits and automatic archiving

The problem we solve

Traditional building site visits are paper-based and time-consuming:

- Inspectors use manual checklists and handwritten notes in the field
- Site visit reports are written up from scratch after each visit
- Documents are manually archived in the municipality's case management system
- Risk of findings going missing, wrong document numbers being used, or archiving being forgotten

This creates extra work for inspectors, delays in case processing, and potential gaps in documentation.

The solution

Tilsynsapp-PNB is a web application that digitalises the entire site visit process — from preparation to archiving — in one seamless workflow.

Preparation (5 min)	→	Field visit (on site)	→	Report generation (automatic)	→	Archive in 360° (one click)
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The inspector works in a browser on PC, tablet, or mobile. When the site visit is complete, the report is generated automatically and archived directly in the municipality's Plan & Building system (Public 360°).

Key features

For the inspector in the field

Feature	Value
Structured checklist	100+ standardised checkpoints, filtered by measure type, property attributes, survey area, and measure type from PNB
Survey area & measure type	Multi-select classification pulled from PNB code tables — controls which checkpoints are displayed
Deviation registration	Mark findings with a comment, responsible contact, and GPS coordinates directly on the checkpoint
Attachments	Upload photos and documents from the site visit, linked to individual checkpoints. The address is automatically resolved from the photo's GPS data and stamped onto the image
Map integration	Pin the site visit location geographically with one tap
Case search	Look up the case from PNB directly in the app — no manual copying or switching between systems

My PNB cases	Dashboard tab showing all PNB cases where the logged-in user is the responsible case officer — with properties, contacts, case stages, and deadlines at a glance
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For the organisation

Feature	Value
Automatic PDF report	Fully formatted site visit report generated without manual effort — includes case metadata, checklist, deviation summary, photos, and map
Archive with one click	Report sent directly to the correct case in Public 360° — create a new document or update an existing one
Traceability	All site visits, deviations, and archival events are logged and searchable
User management	Simple access control via admin panel — no IT department involvement needed
Audit log	All admin actions logged automatically (ISO 27001 A.12.4.1)

Integration with existing systems

The application is built to live **inside** the municipality's existing infrastructure — not alongside it:

- **Public 360° / Plan & Building** — Cases, parties, property, and documents are fetched and stored directly via the SIF API. No duplicate data entry.
- **Municipal login** — Supports Azure AD via OAuth2 (same single sign-on as the rest of the municipality's systems)
- **Browser-based** — No installation required, works on all devices (PC, tablet, mobile)

Benefits

Time savings per inspection

Example: inspection of a new single-family house with 3 deviations

The current process

Step	Description	Time
Preparation	Review relevant documentation, drawings, and case history; look up the case in PNB, note down case number, address, applicant, and measure type	130 min
On site	Carry out the inspection with running notes on a phone or paper; take photos on a phone	60 min
Write report	Open the Word document in PNB, go through notes and photos, write the report, insert photos from the phone	60 min
Archiving	Finalise and dispatch the document in PNB	10 min

Total		260 min
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With Tilsynsapp-PNB

Step	Description	Time
Preparation	Review relevant documentation, drawings, and case history; look up the case in the app — address, applicant, property, and measure type are fetched automatically from PNB	122 min
On site	Carry out the inspection using a structured checklist, record deviations with comments, take photos linked directly to each checkpoint	60 min
Report and archiving	Press "Generate report", press "Archive" — PDF sent directly to the correct case in PNB	2 min
Total		184 min

Result

	Today	With the app	Saved
The inspection itself (unchanged)	60 min	60 min	—
Preparation and follow-up work	200 min	124 min	76 min
Total per inspection	260 min	184 min	76 min (29 %)

A case officer conducting **2 inspections per week** saves around **2.5 hours per week** — equivalent to approximately **100 hours, or 2.5 working weeks, per year**.

Quality and compliance

- Standardised checklists ensure nothing is missed — 100+ checkpoints with legal references (technical regulations, Planning and Building Act)
- Responsible contact can be recorded per deviation for clear accountability
- Complete traceability from inspection to archived document in 360°

Scalability

- Checkpoints and dropdown lists can be configured without development work via the admin panel
- New measure types, inspection areas, and types added through the UI
- Role-based access — separates inspector and administrator responsibilities

Technical platform

The application is built on modern, maintainable technology with no single points of failure:

- **Web application** (Next.js 15 / React 19) — works in all modern browsers, no app installation
- **Database:** Supabase (PostgreSQL) with full encryption and row-level access control (RLS)

- **File storage:** Supabase Storage — inspection attachments stored securely with access control
- **Hosting:** Vercel — automatic scaling, zero server administration, global CDN
- **Security:** HTTPS everywhere, HSTS, CSP headers, audit logging, role-based access, secrets never exposed to browser
- **Code quality:** Biome (linting + formatting), SonarQube integration, and TypeScript strict mode

Performance

The archival flow is optimised for speed through parallel execution:

- SIF case lookup starts immediately alongside database queries
- PDF generation runs in parallel with the pending archival record insert
- File uploads and participant synchronisation run concurrently
- Map image fetch (Kartverket WMS) runs in parallel with attachment downloads

Status and roadmap

Status	Area
✓ Production-ready	Inspection workflow and structured checklists
✓ Production-ready	PDF report generation with inline images, deviation summary, legal references per deviation, and closure deadline (SAK10 § 15-3)
✓ Production-ready	Archiving to Public 360° — create new or update existing document
✓ Production-ready	Auto-dispatch of archived documents to recipients
✓ Production-ready	User and access administration
✓ Production-ready	Audit logging (ISO 27001 A.12.4.1)
⚙️ Configurable	Checkpoint library and dropdown lists customisable per municipality
⚙️ Configurable	SIF archive codes, contact roles, and title template set per installation

Contact and next steps

For a demo, technical walkthrough, or questions about adapting the application to your municipality's setup — contact the project lead.

Tilsynsapp-PNB — developed for Norwegian municipalities with Plan & Building integration